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# Safety data sheet



BASF Safety data sheet according to the United Nations' Globally Harmonized System (UN GHS)
Date / Revised: 20.06.2023 Version: 4.0
Product: **Luphen® 3685**
(ID no. 30626974/SDS_GEN_00/EN)
Date of print 02.07.2025

## 1. Identification

### Product identifier

## Luphen® 3685

### Relevant identified uses of the substance or mixture and uses advised against

Relevant identified uses: Raw material, for industrial use only

### Details of the supplier of the safety data sheet

Company:
BASF SE
67056 Ludwigshafen
GERMANY
Regional Business Unit Dispersions Europe

Telephone: +49 621 60-0
E-mail address: ed-psr@basf.com

### Emergency telephone number

International emergency number:
Telephone: +49 180 2273-112

## 2. Hazards Identification

### Classification of the substance or mixture

According to UN GHS criteria

Skin Sens. 1

For the classifications not written out in full in this section the full text can be found in section 16.

### Label elements

Globally Harmonized System (GHS)

Pictogram:

Signal Word:
Warning

Hazard Statement:

| | |
|---|---|
| H317 | May cause an allergic skin reaction. |

Precautionary Statements (Prevention):

| | |
|---|---|
| P280 | Wear protective gloves. |
| P261 | Avoid breathing dust/fume/gas/mist/vapours/spray. |
| P272 | Contaminated work clothing should not be allowed out of the workplace. |

Precautionary Statements (Response):

| | |
|---|---|
| P302 + P352 | IF ON SKIN: Wash with plenty of soap and water. |
| P333 + P313 | If skin irritation or rash occurs: Get medical attention. |
| P362 + P364 | Take off contaminated clothing and wash it before reuse. |

Precautionary Statements (Disposal):

| | |
|---|---|
| P501 | Dispose of contents and container to hazardous or special waste collection point. |

According to UN GHS criteria

Hazard determining component(s) for labelling: mixture of: 5-chloro-2-methyl-2H-isothiazol-3-one and 2-methyl-2H-isothiazol-3-one

## 3. Composition/Information on Ingredients

### Substances

Not applicable

### Mixtures

Chemical nature

Aqueous dispersion of a polymer based on: polyurethane

Hazardous ingredients (GHS)
According to UN GHS criteria

mixture of: 5-chloro-2-methyl-2H-isothiazol-3-one and 2-methyl-2H-isothiazol-3-one (3:1)

| | |
|---|---|
| Content (W/W): > 0 PPM - < 25 PPM<br>CAS Number: 55965-84-9<br>INDEX-Number: 613-167-00-5 | Acute Tox. 3 (oral)<br>Acute Tox. 2 (Inhalation - mist)<br>Acute Tox. 2 (dermal)<br>Skin Corr./Irrit. 1C<br>Eye Dam./Irrit. 1<br>Skin Sens. 1A<br>Aquatic Acute 1<br>Aquatic Chronic 1<br>M-factor acute: 100<br>M-factor chronic: 100<br>H301, H317, H314, H310 + H330, H400, H410<br>EUH071<br><br><u>Specific concentration limit:</u><br>Skin Sens. 1A: >= 0,0015 %<br>Eye Dam./Irrit. 1: >= 0,6 %<br>Eye Dam./Irrit. 2: 0,06 - < 0,6 %<br>Skin Corr./Irrit. 1C: >= 0,6 %<br>Skin Corr./Irrit. 2: 0,06 - < 0,6 % |

For the classifications not written out in full in this section the full text can be found in section 16.

## 4. First-Aid Measures

### Description of first aid measures

Remove contaminated clothing.

If inhaled:
Keep patient calm, remove to fresh air, seek medical attention.

On skin contact:
Wash thoroughly with soap and water

On contact with eyes:
Immediately wash affected eyes for at least 15 minutes under running water with eyelids held open, consult an eye specialist.

On ingestion:
Rinse mouth and then drink 200-300 ml of water. Do not induce vomiting unless told to by a poison control center or doctor.

### Most important symptoms and effects, both acute and delayed

Symptoms: Information, i.e. additional information on symptoms and effects may be included in the GHS labeling phrases available in Section 2 and in the Toxicological assessments available in Section 11., (Further) symptoms and / or effects are not known so far

Hazards: No hazards anticipated.

### Indication of any immediate medical attention and special treatment needed

Treatment: Symptomatic treatment (decontamination, vital functions).

## 5. Fire-Fighting Measures

### Extinguishing media
Suitable extinguishing media:
water spray, dry powder, foam, carbon dioxide

### Special hazards arising from the substance or mixture

No particular hazards known.

### Advice for fire-fighters
Special protective equipment:
No data available.

Further information:
Dispose of fire debris and contaminated extinguishing water in accordance with official regulations. Product itself is non-combustible; fire extinguishing method of surrounding areas must be considered.

## 6. Accidental Release Measures

### Personal precautions, protective equipment and emergency procedures
Use personal protective clothing. Avoid contact with skin and eyes.

### Environmental precautions
Do not release untreated into natural waters.

### Methods and material for containment and cleaning up
For small amounts: Pick up with suitable absorbent material (e.g. sand, sawdust, general-purpose binder, kieselguhr). Dispose of absorbed material in accordance with regulations.
For large amounts: Pump off product.

## 7. Handling and Storage

### Precautions for safe handling
Handle in accordance with good industrial hygiene and safety practice. After long storage, slight quantities of carbon monoxide may be formed. To our best knowledge, the occupational exposure limit (OEL) is not exceeded during use. Entering of tanks must only be performed after intensive cleaning and when it is ensured that residual vapours have been removed. Consideration of national laws and international standards for confined space entry should be taken in to account. In case of doubt, the concentration of Carbon monoxide must be determined.

### Conditions for safe storage, including any incompatibilities
Further information on storage conditions: Store protected against freezing.

Protect from temperatures below: 5 °C
Protect from temperatures above: 60 °C

### Specific end use(s)
For the relevant identified use(s) listed in Section 1 the advice mentioned in this section 7 is to be observed.

## 8. Exposure Controls/Personal Protection

### Control parameters

Components with occupational exposure limits

55965-84-9: mixture of: 5-chloro-2-methyl-2H-isothiazol-3-one and 2-methyl-2H-isothiazol-3-one

### Exposure controls

Personal protective equipment

Hand protection:
Suitable chemical resistant safety gloves (EN ISO 374-1) also with prolonged, direct contact (Recommended: Protective index 6, corresponding > 480 minutes of permeation time according to EN ISO 374-1): E.g. nitrile rubber (0.4 mm), chloroprene rubber (0.5 mm), butyl rubber (0.7 mm) etc.
Supplementary note: The specifications are based on tests, literature data and information of glove manufacturers or are derived from similar substances by analogy. Due to many conditions (e.g. temperature) it must be considered, that the practical usage of a chemical-protective glove in practice may be much shorter than the permeation time determined through testing.

Eye protection:
Safety glasses with side-shields (frame goggles) (e.g. EN 166)

General safety and hygiene measures
Handle in accordance with good industrial hygiene and safety practice. Wearing of closed work clothing is required additionally to the stated personal protection equipment. Hands and/or face should be washed before breaks and at the end of the shift. Avoid contact with skin and eyes.

## 9. Physical and Chemical Properties

### Information on basic physical and chemical properties

| | | |
|---|---|---|
| Form: | liquid, dispersion | |
| Colour: | white | |
| Odour: | almost odourless | |
| Odour threshold: | not determined | |
| pH value: | 8,0 - 9,0 | (DIN ISO 976) |
| *Information on: Water* | | |
| *Melting point:* | *0 °C* | |
| --- | | |
| *Information on: Water* | | |
| *Boiling point:* | *100 °C* | |
| --- | | |
| Flash point: | not applicable | |
| Flammability: | not flammable | |
| Lower explosion limit: | For liquids not relevant for classification and labelling. | |

| | | |
|---|---|---|
| Upper explosion limit: | For liquids not relevant for classification and labelling. | |
| Ignition temperature: | not applicable | |
| *Information on: Water* | | |
| *Vapour pressure:* | *23,4 hPa*<br>*(20 °C)*<br>*Literature data.* | |
| --- | | |
| Density: | approx. 1,04 g/cm3<br>(20 °C) | (ISO 2811-1) |
| Relative density: | No data available. | |
| Relative vapour density (air): | not applicable | |
| Solubility in water: | partly soluble<br>(15 °C) | |
| Partitioning coefficient n-octanol/water (log Kow): | not applicable | |
| Self ignition: | not self-igniting | |
| Thermal decomposition: | No decomposition if used correctly. | |
| Viscosity, dynamic: | 20 - 120 mPa.s<br>(23 °C, 250 1/s) | (DIN EN ISO 3219, Annex A) |
| Explosion hazard: | not explosive | |
| Fire promoting properties: | not fire-propagating | |

### Other information

| | | |
|---|---|---|
| Self heating ability: | It is not a substance capable of spontaneous heating. | |
| Miscibility with water: | miscible | |
| Solids content: | 39,0 - 41,0 % | (DIN EN ISO 3251) |
| Other Information: | | |
| Range of particle size: < 0,1 µm - 10 µm | | |

## 10. Stability and Reactivity

### Reactivity
No hazardous reactions if stored and handled as prescribed/indicated.

### Chemical stability
The product is stable if stored and handled as prescribed/indicated.

Peroxides: The product does not contain peroxides. The product/the substance has not a tendency towards the formation of peroxide.

### Possibility of hazardous reactions
No hazardous reactions when stored and handled according to instructions. After long storage, slight quantities of carbon monoxide may be formed.

### Conditions to avoid
Avoid extreme temperatures.

### Incompatible materials

Substances to avoid:
No substances known that should be avoided.

### Hazardous decomposition products

:
No hazardous decomposition products if stored and handled as prescribed/indicated.

## 11. Toxicological Information

### Information on toxicological effects

Acute toxicity

Assessment of acute toxicity:
Virtually nontoxic after a single ingestion. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Experimental/calculated data:
LD50 rat (oral): > 2.000 - 10.000 mg/kg

Irritation

Assessment of irritating effects:
If the product adheres to skin, irritation may occur when it dries.

Not irritating to the eyes. Not irritating to the skin. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Experimental/calculated data:
Skin corrosion/irritation rabbit: non-irritant (OECD Guideline 404)

Serious eye damage/irritation rabbit: non-irritant (OECD Guideline 405)

Respiratory/Skin sensitization

Assessment of sensitization:
Skin sensitizing effects were not observed in animal studies. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Germ cell mutagenicity

Assessment of mutagenicity:
The substance was not mutagenic in bacteria. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Carcinogenicity

Assessment of carcinogenicity:
The whole of the information assessable provides no indication of a carcinogenic effect.

Reproductive toxicity

Assessment of reproduction toxicity:
Not expected to cause reproductive toxicity (based on composition).

Developmental toxicity

Assessment of teratogenicity:
The data available for an assessment of the effect of the substance on developmental toxicity are not sufficient for a proper evaluation.

Specific target organ toxicity (single exposure)

Assessment of STOT single:
Based on the available information there is no specific target organ toxicity to be expected after a single exposure.

Repeated dose toxicity and Specific target organ toxicity (repeated exposure)

Assessment of repeated dose toxicity:
No adverse effects were observed after repeated inhalative exposure in animal studies. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Aspiration hazard

not applicable

Other relevant toxicity information

Based on our experience and the information available, no adverse health effects are expected if handled as recommended with suitable precautions for designated uses. The statement was derived from products of similar composition.

## 12. Ecological Information

### Toxicity

Toxicity to fish:
LC50 (96 h) > 100 mg/l, Brachydanio rerio (OECD Guideline 203, static)

Aquatic invertebrates:
EC50 (48 h) > 100 mg/l, Daphnia magna (OECD Guideline 202, part 1, static)

Aquatic plants:
EC50 (72 h) > 100 mg/l, Scenedesmus subspicatus (OECD Guideline 201)
Nominal concentration.

Microorganisms/Effect on activated sludge:

EC20 (0,5 h) > 100 mg/l, activated sludge, domestic (DIN EN ISO 8192-OECD 209-88/302/EEC,P. C)

The inhibition of the degradation activity of activated sludge is not anticipated when introduced to biological treatment plants in appropriate low concentrations.

### Persistence and degradability

Assessment biodegradation and elimination (H2O):
The product can be virtually eliminated from water by abiotic processes e.g. adsorption onto activated sludge.

Elimination information:
> 70 % DOC reduction (OECD 302B; ISO 9888; 88/302/EEC,part C) Easily eliminated from water.

### Bioaccumulative potential

Bioaccumulation potential:
Based on its structural properties, the polymer is not biologically available. Accumulation in organisms is not to be expected.

### Mobility in soil

Assessment transport between environmental compartments:
Volatility: No data available.

### Other adverse effects

No data available.

### Additional information

Adsorbable organically-bound halogen (AOX):
No data available.

Other ecotoxicological advice:
Do not release untreated into natural waters. At the present state of knowledge, no negative ecological effects are expected.

Ecological data are determined by analogy.

## 13. Disposal Considerations

### Waste treatment methods

Must be disposed of or incinerated in accordance with local regulations.

A waste code in accordance with the European waste catalog (EWC) cannot be specified, due to dependence on the usage.
Observe national and local legal requirements.

## 14. Transport Information

**Land transport**

ADR

| | |
|---|---|
| | Not classified as a dangerous good under transport regulations |
| UN number or ID number: | Not applicable |
| UN proper shipping name: | Not applicable |
| Transport hazard class(es): | Not applicable |
| Packing group: | Not applicable |
| Environmental hazards: | Not applicable |
| Special precautions for user | None known |

RID

| | |
|---|---|
| | Not classified as a dangerous good under transport regulations |
| UN number or ID number: | Not applicable |
| UN proper shipping name: | Not applicable |
| Transport hazard class(es): | Not applicable |
| Packing group: | Not applicable |
| Environmental hazards: | Not applicable |
| Special precautions for user | None known |

**Inland waterway transport**

ADN

| | |
|---|---|
| | Not classified as a dangerous good under transport regulations |
| UN number or ID number: | Not applicable |
| UN proper shipping name: | Not applicable |
| Transport hazard class(es): | Not applicable |
| Packing group: | Not applicable |
| Environmental hazards: | Not applicable |
| Special precautions for user: | None known |

Transport in inland waterway vessel
Not evaluated

**Sea transport**

IMDG

| | |
|---|---|
| | Not classified as a dangerous good under transport regulations |
| UN number or ID number: | Not applicable |
| UN proper shipping name: | Not applicable |
| Transport hazard class(es): | Not applicable |
| Packing group: | Not applicable |
| Environmental hazards: | Not applicable |
| Special precautions for | None known |

user

**Air transport**

IATA/ICAO

| | |
|---|---|
| | Not classified as a dangerous good under transport regulations |
| UN number or ID number | Not applicable |
| UN proper shipping name: | Not applicable |
| Transport hazard class(es): | Not applicable |
| Packing group: | Not applicable |
| Environmental hazards: | Not applicable |
| Special precautions for user | None known |

### Maritime transport in bulk according to IMO instruments

Maritime transport in bulk is not intended.

## 15. Regulatory Information

### Safety, health and environmental regulations/legislation specific for the substance or mixture

Not applicable

## 16. Other Information

Any other intended applications should be discussed with the manufacturer.

Full text of classifications, hazard symbols and hazard statements, if mentioned in section 2 or 3:

| | |
|---|---|
| Skin Sens. | Skin sensitization |
| Acute Tox. | Acute toxicity |
| Skin Corr./Irrit. | Skin corrosion/irritation |
| Eye Dam./Irrit. | Serious eye damage/eye irritation |
| Aquatic Acute | Hazardous to the aquatic environment - acute |
| Aquatic Chronic | Hazardous to the aquatic environment - chronic |
| H301 | Toxic if swallowed. |
| H317 | May cause an allergic skin reaction. |
| H314 | Causes severe skin burns and eye damage. |
| H310 + H330 | Fatal in contact with skin or if inhaled. |
| H400 | Very toxic to aquatic life. |
| H410 | Very toxic to aquatic life with long lasting effects. |
| EUH071 | Corrosive to the respiratory tract. |

The data contained in this safety data sheet are based on our current knowledge and experience and describe the product only with regard to safety requirements. This safety data sheet is neither a Certificate of Analysis (CoA) nor technical data sheet and shall not be mistaken for a specification agreement. Identified uses in this safety data sheet do neither represent an agreement on the corresponding contractual quality of the substance/mixture nor a contractually designated use. It is the

responsibility of the recipient of the product to ensure any proprietary rights and existing laws and legislation are observed.

Vertical lines in the left hand margin indicate an amendment from the previous version.